

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# RATIO, (BEST VALUE PROBLEMS)

**LO: I can compare products using unit pricing to determine the best value.**

## Best value

A **best value** problem asks you to compare products by finding the **price per unit** (per gram, per ml, per item) or the **amount per pound**.

**Method:** Divide the price by the quantity (or vice versa) to find a common rate for comparison.

## Quick examples

|   |                                     |                       |
|---|-------------------------------------|-----------------------|
| ● | 500 g for £2.00 = 0.4p/g            | price per gram        |
| ● | 750 ml for £1.50 = 0.2p/ml          | price per ml          |
| ● | Lower price per unit = better value | compare rates         |
| ● | Bigger pack is always better value  | not always true       |
| ● | Cheapest total price = best value   | must compare per unit |

## Key idea

Best value = lowest price per unit. Divide price by quantity to compare.

$$\text{Price per unit} = \frac{\text{Price}}{\text{Quantity}} \div$$

**The lowest rate is the best deal**

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 – price per gram

Pack A: 400 g for £2.80. Pack B: 600 g for £3.90. Which is better value?

$$\text{A: } 280\text{p} \div 400 = 0.70\text{p per gram}$$

$$\text{B: } 390\text{p} \div 600 = 0.65\text{p per gram}$$

Pack B is cheaper per gram

**Pack B (0.65p/g vs 0.70p/g)**

Compare the price per gram to find the best deal.

### Example 2 – price per item

6-pack of juice for £2.70 or 10-pack for £4.20. Best value?

$$\text{6-pack: } 270\text{p} \div 6 = 45\text{p each}$$

$$\text{10-pack: } 420\text{p} \div 10 = 42\text{p each}$$

10-pack is better value

**10-pack (42p vs 45p each)**

Divide total price by number of items.

### Example 3 – amount per pound

Shop A: 250 g of cheese for £2.50.  
Shop B: 300 g for £2.70.

$$\text{A: } 250 \div 2.50 = 100 \text{ g per pound}$$

$$\text{B: } 300 \div 2.70 = 111.1 \text{ g per pound}$$

Shop B gives more per pound

**Shop B (111 g/£ vs 100 g/£)**

You can also compare grams per pound.

### Example 4 – with offers

Normal: £1.50 for 330 ml. Offer: 3 for £4.00 (330 ml each). Better value?

$$\text{Normal: } 150\text{p} \div 330 = 0.45\text{p/ml}$$

$$\text{Offer: } 400\text{p} \div 990 = 0.40\text{p/ml}$$

Offer is better value

**Offer is better (0.40p/ml vs 0.45p/ml)**

Calculate the total volume in the offer, then find price per ml.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Price per gram

Find the price per gram and state which is better value.

- a) 200 g for £1.60 or 350 g for £2.45
- b) 500 g for £3.00 or 1.2 kg for £6.60
- c) 150 g for £1.80 or 400 g for £4.40
- d) 750 g for £4.50 or 1 kg for £5.80

### 2. Price per ml or litre

Find the price per ml and state which is better value.

- a) 500 ml for £1.20 or 1.5 L for £3.00
- b) 330 ml for £0.89 or 2 L for £4.50
- c) 750 ml for £1.95 or 1 L for £2.40
- d) 250 ml for £0.75 or 400 ml for £1.08

### 3. Price per item

Find the price per item.

- a) 4-pack for £3.20 or 6-pack for £4.50
- b) 8 rolls for £3.60 or 12 rolls for £4.80
- c) Box of 10 for £5.50 or box of 15 for £7.80
- d) 3 for £2.25 or 5 for £3.50

### 4. With special offers

Compare the standard price with the offer.

- a) £2.00 each or 3 for £5.00
- b) £1.50 for 500 ml or buy 2 get 1 free at £1.50
- c) £3.00 for 400 g or 25% extra free (500 g) for £3.00
- d) £4.50 for 6 or £8.00 for 12

### 5. Mixed best-value problems

Solve each problem.

a) Juice: 1 L for £1.10 or 1.5 L for £1.65    b) Cereal: 500 g for £2.40 or 750 g for £3.60    c) Petrol: 40 L for £58 or 25 L for £36    d) Internet: £25/month for 50 GB or £35/month for 100 GB

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Best value means lowest price per unit ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) The cheapest total price is always the best value ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) To compare, find price ÷ quantity for each ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) Bigger packs are always better value ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) You can compare grams per pound or price per gram ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A supermarket sells orange juice in 3 sizes: **Small 330 ml £0.85, Medium 1 L £2.20, Large 2 L £3.90**. Which is the best value? A family needs 4 L per week. What is the cheapest combination?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) Brand A washing powder costs **£4.50 for 30 washes**. Brand B costs **£7.20 for 50 washes**. Brand A has an offer: buy 2 get 1 free. Which is the cheapest per wash with the offer applied?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

